Aryan Patel

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RESEARCH INTERESTS

Deep Learning Systems & Architectures · Foundation Models · Scientific Computing · Data Analysis

EDUCATION

Jabalpur Engineering College — Jabalpur, MP
Bachelor of Technology – Artificial Intelligence and Data Science — *Aug 2025 – Jun 2029*
Standing: Second Year (Sophomore)

Stemfield International School — Jabalpur, MP
Higher Secondary Education – PCM + Computer Science (CS) — *Apr 2011 – Mar 2025*

RESEARCH PROJECTS

NeuralFabric: Machine Learning Framework from Scratch | *Python, NumPy*
- Built a modular machine learning framework from first principles using Python and NumPy, implementing tensor operations, reverse-mode automatic differentiation, and foundational regression models.
- Developed numerical-gradient tests with Pytest to validate autograd correctness, enforced static typing with MyPy, and structured the project for open-source distribution with modern Python packaging and GitHub Actions.

AdityaL1-MAG: Solar Magnetic Field Analysis | *Python, Signal Processing*
- Developed a data-analysis pipeline for ISRO's Aditya-L1 MAG observations, including data ingestion, filtering, and visualization of spacecraft magnetic-field measurements.
- Performed magnetic-field characterization and exploratory analysis to identify patterns and variations in the observed environment.

Timeless: Decoder-Only Transformer Language Model | *PyTorch, Transformers*
- Built a decoder-only Transformer language model from scratch in PyTorch, implementing tokenization, causal self-attention, multi-head attention, residual connections, and normalization.
- Developed an end-to-end LLM workflow covering pretraining on unlabeled text, model checkpointing, text-classification fine-tuning, and instruction tuning for downstream tasks.

Diagnostic Neural Network for Oncology | *TensorFlow, Scikit-learn*
- Built a TensorFlow neural network for breast cancer classification using standardized clinical features and a structured preprocessing pipeline with Scikit-learn.
- Evaluated model performance using accuracy, confusion matrix, and classification metrics, achieving ∼94% test accuracy, and saved the trained model for reuse.

ADDITIONAL ENGINEERING PROJECTS

Kryptika: Blockchain in Pure Python | *ECDSA, P2P, HTTP*
- Implemented a blockchain from scratch in pure Python, covering SHA-256 hashing, Proof of Work, ECDSA-signed transactions, transaction validation, and longest-chain consensus.

DoorDarshan: Real-Time Video Conferencing Platform | *FastAPI, WebSocket, WebRTC*
- Engineered a real-time video conferencing platform using FastAPI, WebSockets, and browser-native WebRTC for live audio/video communication, screen sharing, and peer signaling.

CERTIFICATIONS

Machine Learning Specialization | *DeepLearning.AI* — Jul 2026
- Completed coursework covering supervised learning, neural networks, model optimization, and regularization.

AWARDS & HONOURS

Finalist, RAGNAROK Hackathon | *Google Developer Groups (GDG)* — Jan 2026
- Selected as a finalist in a competitive hackathon for technical problem-solving and rapid project development.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Go, R, JavaScript, TypeScript
Machine Learning: PyTorch, TensorFlow, XGBoost, Scikit-learn
Scientific Computing: NumPy, Pandas, Matplotlib, Scientific Data Analysis
Engineering: Linux, Bash, Git, GitHub